

Lec 7:

- 1) involves executing the system with test cases that are derived from the specification of the real data to be processed by the system.
 - a) Validation
 - b) Verification
 - c) System testing
- 2)the software should conform to its specification.
 - a) Validation
 - b) Verification
 - c) System testing
- 3) ...the software should do what the user really requires.
 - a) Validation
 - b) Verification
 - c) System testing
- 4) ... Concerned with analysis of the static system representation to discover problems (static verification).
 - a) System testing
 - b) Software inspections
 - c) Validation
- 5) Concerned with exercising and observing product behaviour (dynamic verification).
 - a) Software testing
 - b) Software inspections
 - c) Validation
- 6) which of the following is correct about Software inspections:
 - a) Do not require execution of a system .
 - b) Very effective technique for discovering errors.
 - c) May be applied to any representation of the system.
 - d) All of the above.
- 7) ... Involve people examining the source representation with the aim of discovering anomalies and defects.
 - a) System testing
 - b) Software inspections
 - c) Validation

- 8) ... who finds errors, omissions and inconsistencies.
a) Inspector
b) Reader
c) Moderator
- 9) ... who chairs the meeting and notes discovered errors.
d) Inspector
e) Reader
f) Moderator
- 10) Inspections and testing are complementary and not opposing verification techniques. (T / F)
- 11) inspections can check non-functional characteristics such as performance, usability. (T / F) (cannot)
- 12) Inspections can check conformance with a specification but not conformance with the customer's real requirements . (T / F) .
- 13) Inspections and testing should be used during the V&V process .(T/F)
- 14) .. the process of executing a program with the intention of finding errors.
g) Inspector
h) Testing
i) Moderator
- 15) can show the presence of bugs but never their absence.
a) Inspector
b) Testing
c) Moderator
- 16) Which of the following is one of program testing goal:
a) demonstrate to the developer and the customer that the software meets its requirements
b) discover situations in which the behavior of the software is incorrect, undesirable or does not conform to its specification.
c) Both a and b.
- 17) successful test is one that shows that a requirements has been properly implemented.
a) Validation testing
b) Defect testing
c) Verification
- 18) discover system defects.
a) Validation testing

b) Defect testing

c) Verification

19) also known as a **bug**, is a result of an error which can cause system to fail:

a) Error

b) Fault

c) Failure

20) is said to be the **inability** of the system to perform the desired task:

a) Error

b) Fault

c) Failure

21) occurs when fault exists in the system.

d) Error

e) Fault

f) Failure

22) specifications of the **inputs** to the test and the expected **output** from the system:

a) Test case

b) Test data

c) Test Suite

23) a set of test cases for a system:

d) Test case

e) Test data

f) Test Suite

24) are the inputs that have been **devised** to test a system.

g) Test case

h) Test data

i) Test Suite

a. **stories** to be tested:

j) Test case

k) Test scenarios

l) Test Suite

25) where the system is tested during **development** to discover **bugs** and **defects**:

a) Development testing

b) Release testing

c) User testing

26) where a separate testing team test a **complete version** of the system before it is released to users:

d) Development testing

e) Release testing

f) User testing

27) focus on testing the functionality of **objects or methods** :

- a) Unit testing
 - b) Component testing
 - c) System testing
- 28) where several individual units are integrated to create composite components:
- d) Unit testing
 - e) Component testing
 - f) System testing
- 29) focus on testing component interactions:
- g) Unit testing
 - h) Component testing
 - i) System testing
- 30) focus on testing component interfaces:
- a. Unit testing
 - j) Component testing
 - k) System testing
- 31) It is a defect testing process:
- l) Unit testing
 - m) Component testing
 - n) System testing
- 32) Units may be object classes with several attributes and methods. (T / F)
- 33) Unit testing frameworks provide generic test classes that you extend to create specific test cases. (T / F)
- 34) In automated unit testing, you make use of a test automation framework (such as JUnit) to write and run your program tests. (T / F)
- 35) initialize the system with the test case, namely the inputs and expected outputs:
- a) setup part
 - b) call part
 - c) assertion part
- 36) where you call the object or method to be tested :
- d) setup part
 - e) call part
 - f) assertion part
- 37) where you compare the result of the call with the expected result:
- g) setup part
 - h) call part
 - i) assertion part
- 38) where you identify groups of inputs that have common characteristics and should be processed in the same way :
- a) Partition testing
 - b) Guideline-based testing
- 39) where you use testing guidelines to choose test cases :
- c) Partition testing
 - d) Guideline-based testing
- 40) a black-box test technique to reduce # of required test cases :

- a) Equivalence partitioning
 - b) Guideline-based testing
- 41) where users or potential users of a system test the system in their own environment:
- g) Development testing
 - h) Release testing
 - i) User testing
-

Lec 8:

- 1) is the process of testing a particular release of a system that is intended for use outside of the development team:
 - a) unit testing
 - b) Release testing
 - c) User testing
- 2) Release testing has to show that the system delivers its specified functionality, performance and dependability, and that it does not fail during normal use.
 - a) True
 - b) False
- 3) Release testing is usually a black-box testing process where tests are only derived from the system specification :
 - a) True
 - b) False
- 4) Release testing is a form of system testing. (T / F)
- 5) System testing by the development team should focus on discovering bugs in the system
 - a) True
 - b) False
- 6) users of the software work with the development team to test the software at the developer's site:
 - a) Alpha testing
 - b) Beta testing
 - c) Acceptance testing
- 7) Similar to beta testing, but the main purpose of it is to get final user acceptance/approval of system to be delivered:
 - d) Alpha testing
 - e) Beta testing
 - f) Acceptance testing
- 8) It serves as a live application of the software in an environment that cannot be controlled by the developer :
 - g) Alpha testing
 - h) Beta testing
 - i) Acceptance testing
- 9) In ... the end-user records all problems that are encountered and reports these to the developers at regular intervals:

- j) Alpha testing
 - k) Beta testing
 - l) Acceptance testing
- 10) Black-box tests are driven by the program's specification. (T / F)
- 11) White-box tests are driven by the program's implementation. (T/F)
- 12) Performance testing is a non-functional testing technique. (T/ F)
- 13) Performance testing measures the quality attributes of the system, such as :
- a) scalability
 - b) reliability
 - c) resource usage
 - d) All of the above
- 14) checks the application's ability to perform under anticipated user loads
- a) Load testing
 - b) Stress testing
- 15) The objective is to identify the breaking point of an application :
- c) Load testing
 - d) Stress testing
- 16) involves testing an application under extreme workloads:
- e) Load testing
 - f) Stress testing
- 17) is a measure of how responsive an application or subsystem is to a client request
- a) Response time
 - b) Scalability
 - c) Speed
- 18) Determines whether the application responds quickly
- d) Response time
 - e) Scalability
 - f) Speed
- 19) Determines maximum user load the software application can handle
- g) Response time
 - h) Scalability
 - i) Speed
- 20) Determines if the application is stable under varying loads
- j) Response time
 - k) Speed
 - l) Stability
- 21) is a measure which describes the degree of which the source code of the program has been tested
- a) Test coverage
 - b) Statement coverage
- 22) reports the true or false outcomes of each Boolean expression
- c) Test coverage
 - d) Decision coverage
- 23) also named open-box, clear box, glass box:
- a) White-box testing
 - b) Black-box testing

- 24) is the number of units of work that can be handled per unit of time :
- m) Response time
 - n) Throughput
 - o) Speed
- 25) checking that it does what the customer wants
- a) Specification
 - b) Design and implementation
 - c) Evolution
 - d) Validation
- 26) defining what the system should do
- e) Specification
 - f) Design and implementation
 - g) Evolution
 - h) Validation
- 27) defining the organization of the system and implementing the system
- i) Specification
 - j) Design and implementation
 - k) Evolution
 - l) Validation
- 28) changing the system in response to changing customer needs.
- m) Specification
 - n) Design and implementation
 - o) Evolution
 - p) Validation
- 29) Software evolution processes depend on:
- a) The type of software being maintained
 - b) The development processes used
 - c) The skills and experience of the people involved.
 - d) All of the above
- 30) Maintenance does not normally involve major changes to the system's architecture:
- a) True
 - b) False
- 31) Adding or modifying the system's functionality to meet new requirements
- a) Perfective maintenance
 - b) Adaptive maintenance
 - c) Corrective maintenance
- 32) Changing a system to adapt it to new hardware or operating system
- d) Perfective maintenance
 - e) Adaptive maintenance
 - f) Corrective maintenance
- 33) Changing a system to fix coding, design, or requirements errors
- g) Perfective maintenance

- h) Adaptive maintenance
 - i) Corrective maintenance
- 34) Maintenance costs are reduced if the same staff are involved with them for some time:
- a) Team stability
 - b) Contractual responsibility
 - c) Staff skills
 - d) Program age and structure
- 35) The developers of a system may have no contractual responsibility for maintenance so there is no incentive to design for future change
- e) Team stability
 - f) Contractual responsibility
 - g) Staff skills
 - h) Program age and structure
- 36) Maintenance staff are often inexperienced and have limited domain knowledge
- i) Team stability
 - j) Contractual responsibility
 - k) Staff skills
 - l) Program age and structure
- 37) as programs age, their structure is degraded and they become harder to understand and change:
- m) Team stability
 - n) Contractual responsibility
 - o) Staff skills
 - p) Program age and structure
- 38) To make legacy software systems easier to maintain, you can reengineer these systems to improve their structure and understandability . (T /F)
- 39) Reengineering may involve:
- a) redocumenting the system
 - b) refactoring the system architecture
 - c) Both a and b
- 40) Reengineering may involve:
- a) translating programs to a modern programming language
 - b) modifying and updating the structure and values of the system's data
 - c) Both a and b
- 41) Which of the following is one of reengineering process activities:
- a) Source code translation
 - b) Reverse engineering
 - c) Program structure improvement
 - d) All of them
- 42) Restructure automatically for understandability:
- e) Source code translation
 - f) Reverse engineering
 - g) Program structure improvement
- 43) Analyse the program to understand it ,This helps to document its organization and functionality :
- h) Source code translation
 - i) Reverse engineering

- j) Program structure improvement
 - 44) Related parts of the program are grouped together, and where appropriate, redundancy is removed, in some cases this stage may involve architectural refactoring
 - k) Source code translation
 - l) Program structure improvement
 - m) Program modularization
 - 45) clean up the data by finding and correcting mistakes, removing duplicate records:
 - n) Program structure improvement
 - o) Program modularization
 - p) Data reengineering
-

Lec 9:

- 1) The probability that the system will be up and running and able to deliver useful services to users:
 - a) Availability
 - b) Reliability
 - c) Safety
 - d) Security
- 2) The probability that the system will correctly deliver services as expected by users:
 - e) Availability
 - f) Reliability
 - g) Safety
 - h) Security
- 3) A judgment of how likely it is that the system will cause damage to people or its environment:
 - i) Availability
 - j) Reliability
 - k) Safety
 - l) Security
- 4) A judgment of how likely it is that the system can resist accidental or deliberate intrusions:
 - m) Availability
 - n) Reliability
 - o) Safety
 - p) Security
- 5) A system may be unreliable because its data has been corrupted by an external attack
 - a) True
 - b) False
- 6) Unauthorized users cannot read information:
 - a) Confidentiality
 - b) Integrity
 - c) Availability
- 7) Unauthorized users cannot alter information:
 - d) Confidentiality

- e) Integrity
- f) Availability
- 8) Authorized users can always access information:
 - g) Confidentiality
 - h) Integrity
 - i) Availability
- 9) A worm that deletes data in a system is an example of:
 - j) Confidentiality
 - k) Integrity
 - l) Availability
- 10) A denial-of-service attack that overloads a server is an example of:
 - m) Confidentiality
 - n) Integrity
 - o) Availability
- 11) the theft of credit card data from an e-commerce system is an example of:
 - p) Confidentiality
 - q) Integrity
 - r) Availability
- 12) Data in its encrypted form:
 - a) Encryption
 - b) Plaintext
 - c) Ciphertext
- 13) The transformation of information using a secret:
 - d) Encryption
 - e) Plaintext
 - f) Ciphertext
- 14) Data input to the Cipher or output from the Inverse Cipher:
 - g) Encryption
 - h) Plaintext
 - i) Ciphertext
- 15) Something of value which has to be protected, It may be the software system itself or data used by that system:
 - a) Asset
 - b) Vulnerability
 - c) Attack
- 16) A weakness in a computer-based that may be exploited to cause loss or harm:
 - d) Asset
 - e) Vulnerability
 - f) Attack
- 17) An exploitation of a system's vulnerability. Generally, this is from outside the system and is a deliberate attempt to cause some damage:
 - g) Asset
 - h) Vulnerability
 - i) Attack
- 18) Possible loss or harm to a computing system:
 - j) Asset
 - k) Exposure

- l) Attack
- 19) protective measure that reduces a system's vulnerability :
 - m) Asset
 - n) Control (countermeasures)
 - o) Attack
- 20) Circumstances that have potential to cause loss or harm, you can think of these as system vulnerability that is subjected to an attack:
 - p) Control (countermeasures)
 - q) Attack
 - r) Threat
- 21) allow an attacker to gain access to an asset
 - a) Fabrication threats
 - b) Modification threats
 - c) Interruption threats
 - d) Interception threats
- 22) allow an attacker to make part of the system unavailable
 - e) Fabrication threats
 - f) Modification threats
 - g) Interruption threats
 - h) Interception threats
- 23) that allow an attacker to insert false information into a system
 - i) Fabrication threats
 - j) Modification threats
 - k) Interruption threats
 - l) Interception threats
- 24) allow an attacker to tamper with a system asset
 - m) Fabrication threats
 - n) Modification threats
 - o) Interruption threats
 - p) Interception threats
- 25) The system is forced into a state where normal services are unavailable or where service provision is significantly degraded:
 - a) Denial of service
 - b) Corruption of programs or data
 - c) Disclosure of confidential information
- 26) Information that is managed by the system may be exposed to people who are not authorised to read or use that information:
 - d) Denial of service
 - e) Corruption of programs or data
 - f) Disclosure of confidential information
- 27) The programs or data in the system may be modified in an unauthorised way:
 - g) Denial of service
 - h) Corruption of programs or data
 - i) Disclosure of confidential information
- 28) specify whether or not a system should identify its users before interacting with them
 - a) Authentication requirements

- b) Authorization requirements
 - c) Identification requirements
 - d) Integrity requirements
- 29) specify how data corruption can be avoided
- e) Authentication requirements
 - f) Authorization requirements
 - g) Identification requirements
 - h) Integrity requirements
- 30) specify how a system should protect itself against viruses, worms, and similar threats:
- i) Authentication requirements
 - j) Authorization requirements
 - k) Identification requirements
 - l) Immunity requirements
- 31) specify the privileges and access permissions of identified users
- m) Authentication requirements
 - n) Authorization requirements
 - o) Identification requirements
 - p) Integrity requirements
- 32) specify how users are identified
- q) Authentication requirements
 - r) Authorization requirements
 - s) Identification requirements
 - t) Integrity requirements
- 33) requirements specify what mechanisms should be used to detect attacks on the system:
- a) Privacy requirements
 - b) Non-repudiation requirements
 - c) Intrusion detection
- 34) specify that a party in a transaction cannot deny its involvement in that transaction:
- d) Privacy requirements
 - e) Non-repudiation requirements
 - f) Intrusion detection
- 35) specify how data privacy is to be maintained
- g) Privacy requirements
 - h) Non-repudiation requirements
 - i) Intrusion detection
- 36) Identify the key system assets (or services) that have to be protected:
- a) Asset identification
 - b) Asset value assessment
 - c) Exposure assessment
 - d) Threat identification
- 37) Estimate the value of the identified assets:
- e) Asset identification
 - f) Asset value assessment
 - g) Exposure assessment
 - h) Threat identification
- 38) Assess the potential losses associated with each asset:

- i) Asset identification
 - j) Asset value assessment
 - k) Exposure assessment
 - l) Threat identification
- 39) Identify the most probable threats to the system assets:
- m) Asset identification
 - n) Asset value assessment
 - o) Exposure assessment
 - p) Threat identification
- 40) Decompose threats into possible attacks on the system and the ways that these may occur:
- a) Attack assessment
 - b) Control identification
 - c) Feasibility assessment
 - d) Security requirements definition
- 41) Propose the controls that may be put in place to protect an asset:
- e) Attack assessment
 - f) Control identification
 - g) Feasibility assessment
 - h) Security requirements definition
- 42) Assess the technical feasibility and cost of the controls:
- i) Attack assessment
 - j) Control identification
 - k) Feasibility assessment
 - l) Security requirements definition
- 43) Define system security requirements these can be infrastructure or application system requirements:
- m) Attack assessment
 - n) Control identification
 - o) Feasibility assessment
 - p) Security requirements definition

Shift

Shift